

Eurasian woodcock (*Scolopax rusticola*) in intensively managed Central European forests use large home ranges with diverse habitats

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ABSTRACT

Modern management of Europe's forests should strive for a delicate balance between efficient production, which is their primary purpose, and forest management practices that conserve and support biodiversity. This requires detailed knowledge of the ecology of forest-dwelling species, including their site-specific behavior and habitat preferences. One such species is the Eurasian woodcock (*Scolopax rusticola*), an important game bird that breeds throughout the Palearctic region. Despite its importance, our knowledge of woodcock ecology is limited, and collecting reliable data is challenging due to the species' cryptic behavior. Here, we used high-precision GPS-GSM telemetry to determine home range size and to investigate habitat selection in male Eurasian woodcocks inhabiting Central European forests. Using Bayesian mixed-effects models corrected for spatial autocorrelation, we compared several habitat characteristics (including dominant tree and herb species) of sites where woodcocks occurred with randomly selected sites within their home ranges. Our results show that woodcock in Central Europe move over much larger home ranges during the breeding season (median 7.26 km²) than previously reported in other parts of their breeding range, although they regularly use only small part of this large area (median 1.09 km²). Within their home range, woodcocks used a wide range of habitats, including intensively managed conifer monocultures, but preferred deciduous forests, particularly those dominated by European white birch (*Betula pendula*). Specifically, 24 % of the plots with woodcock were dominated by birch, compared to only 12 % of the randomly selected plots. Importantly, regardless of dominant tree species, woodcocks preferred sites with dense vegetation cover, but good terrain traversability. During the day, woodcocks preferred sites with dense canopy cover, while at night they preferred more open areas.

However, in contrast to previously published results elsewhere, these open areas were more likely to be sparse forest (73%) than grasslands or non-forest cropland (24%), and were characterized by less continuous canopy cover, with a median of only 40%.

Our study highlights the importance of specific microhabitats, such as sites with dominant representation of commercially non-target broadleaved species (birch), or sites with sparse canopy cover but denser understorey vegetation. Appropriate management practices that promote such specific microhabitats would have a positive impact on populations of woodcock and other forest species inhabiting commercial forests in Central Europe.

1. Introduction

Across Europe, the intensive production forestry has led to large scale changes in the forest structure and species composition, namely the simplification of the forest age structure, creating monocultures, and preference for planting coniferous trees (Fanta 1997, Hartley 2002). Such changes often had negative impact on biodiversity (Felton et al. 2010; Felton et al., 2011). Forests in Central Europe are further changing

due to the climate change and accompanying phenomena, such as a drought (Schuldt et al. 2020), windstorms (Schelhaas et al. 2010), or eight-toothed spruce bark beetle (*Ips typographus*) calamities (Hlásny et al. 2021). All of this can have substantial, albeit not necessarily negative impact on biodiversity (Archaux and Wolters 2006, Beudert et al. 2015, Kebrle et al. 2022). Consequently, many authors call for replacing monocultures by more diverse forests and therefore another broadscale change in European forests can be expected in future

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(Hartley 2002, Knoke et al. 2008, Felton et al. 2010). Detailed knowledge of ecology of forest inhabiting species is thus crucial for making informed and impactful decisions.

Eurasian woodcock (*Scolopax rusticola*) is an iconic and popular game bird, breeding across the Palearctic region (Cramp and Simmons 1983). Today, the woodcock is considered one of the most important game birds, being frequently hunted in most of European countries, with about 2.5–3.5 million individuals harvested each year (Hirschfeld and Heyd 2005, Christensen et al. 2017). The woodcock population has been shown to decline in some countries (Machado et al. 2008, Heward et al. 2015), while considered stable in others (BirdLife International 2023). However, due to a very specific ecology, even basic data about the woodcock local distribution and abundance are usually of poor quality (Christensen et al. 2017, BirdLife International 2023). Unfortunately, the collection of reliable and robust data about the specific aspects of woodcock ecology, such as seasonal variation in habitat use, or the size of home range is challenging.

The main problem is that the woodcocks are cryptic, being easily detectable only during a short roding period around the dusk (Cramp and Simmons 1983). Counting of the roding males and even recording the habitat characteristics of the roding sites became a popular method of studying woodcocks and their habitat preferences (Machado et al. 2008, Braña et al. 2013). However, given that an individual woodcock male may use a much larger area (Hirons 1980), such data can only provide a very coarse information about the preferred habitats. Getting an unbiased insight into the true habitat use and preferences of woodcocks outside the roding period is only possible using telemetry.

Previous work suggests that woodcocks inhabit a very broad scale of forest habitats, with preferences for habitats with high abundance of earthworms (Hirons and Johnson 1986, Duriez and Corda, 2005, Hoodless and Hirons 2007, Braña et al. 2013), their good accessibility (e. g. places with shallow litter: Hirons and Johnson 1986), and places where woodcocks are safe from being predated (i.e. places covered by dense vegetation Hirons and Johnson 1986, Duriez and Corda, 2005). Several studies also reported that woodcocks tend to change their location between day and night, with a preference for open habitats at night (particularly pastures) and forests during the day, mainly during wintering (Duriez et al., 2005a, Hoodless and Hirons 2007, Braña et al. 2010). However, the data about the behavior and habitat preferences are still very limited, constrained mostly to the wintering sites in southern and western Europe (Duriez et al., 2005a; Duriez et al., 2005b; Braña et al. 2010), or specific areas at the edge of the species breeding range (Hoodless and Coulson 1998, Hoodless and Hirons 2007, Machado et al. 2008, Braña et al. 2013). These life history and habitat use aspects remain unstudied in central and eastern European forests, which host an important part of the woodcock population being hunted (Hobson et al. 2013). Given the diversity of forests and forest management practices across the entire woodcock breeding range, it may be difficult to generalize results from a single site. For example, the British woodcock's preference for dense cover of dog's mercury (*Mercurialis perennis*) (Hirons and Johnson 1986) is not widely applicable across the Europe because of the limited distribution of forests where dog's mercury predominates among the herbs. Therefore, conducting additional studies from areas with different forest composition is necessary to gain a comprehensive understanding of woodcock habitat preferences and use. Finally, the number of analyzed locations per individual in published studies is usually relatively small (but see: Hoodless and Hirons 2007) and there is so far only one study, quantifying the habitat use based on the continuous tracking of particular individuals throughout the whole breeding season (Hoodless and Hirons 2007).

Here, we analyzed the habitat and home range size of a Central European Eurasian Woodcock population inhabiting diverse forests, predominantly commercially harvested forests, but interspersed with patches of early successional and deciduous forests, which we expected to be preferred by woodcocks. By tracking males captured during the roding period and outfitted using a high-precision GPS-GSM

datalogger, we: 1) calculated the home ranges of individual males and described the frequency and length of movements with respect to the time of day; 2) provided a general habitat classification of the sites where the GPS position was recorded; and 3) compared the characteristics of the sites visited by the woodcock with the randomly selected plots within the woodcocks home ranges. This allowed us to assess habitat selection (preference/avoidance) for a wide range of habitat structure features, as well as dominant tree and herb species. Finally, we compared the habitat preferences between the day and night, and throughout the breeding season.

In generally, we tested the following three main predictions. Firstly, we expected woodcocks to prefer sites with good traversability but also with sufficient shelter. Second, we expected them to prefer forest stands with a high proportion of deciduous trees, especially early successional species such as birch (*Betula* sp.). Thirdly, we tested whether woodcocks would avoid intensively managed conifer monocultures. We discuss our findings in the context of the current forest structure and plant species composition of Central European forests, which may be useful for improving diversity-promoting management of commercial forests.

2. Methods

2.1. Study areas

Three study areas in the Czechia were selected to cover different Central European geographical regions and forest types. All three areas consist of a large and continuous forest complexes and are known to host a significant population of Eurasian woodcocks (Šťastný et al. 2021).

2.1.1. Ralsko

This former military area is located in the Liberec region, northern Bohemia. It covers more than 300 km² of continuous forest, mostly between 270 and 300 m a.s.l., with only a few volcanic hills rising above 500 m a.s.l. The region is formed by Cretaceous sediments with sand deposits and is characterized by poor soil fertility (Chytrý, 2012; Czech Geological Survey, 2023). The region experiences a moderately warm and dry climate (Vondráková et al., 2013) with an average air temperature of 7.2 °C during the breeding season (April – August; CHMI, 2023) and average annual precipitation of 635 mm. The forests are a mixture of intensively managed coniferous monocultures and mixed/deciduous forests, often with a natural character (i.e. extensively managed, with diversified age structure and without harvesting of fallen trees). In general, the forests are dominated by Scots pine (*Pinus sylvestris*) and Norway spruce (*Picea abies*), but also by beech (*Fagus sylvatica*), birch (*Betula pendula*) or oak (*Quercus petraea* and *Q. robur*) (Novák et al. 2012). The dominance of Scots pine is determined by the mostly sandy soils. Approximately half of the forest complex has been declared a Special Protection Area under the EU Wild Birds Directive and a Protected Landscape Area under Czech legislation (<https://drusop.nature.cz/ost/chrobjekty/ptacob>). Several parts of the area have also been declared as nature reserves. We captured woodcocks in two different locations, about 8 km apart: Vrchbělá (50.53 N, 14.75 E) and the former Hradčany military airport (50.62 N, 14.75 E). Vrchbělá is mostly covered by forests dominated by Scots pine and birch, with substantial areas left to natural succession in various stages. Hradčany airport is surrounded by a variety of habitats, ranging from artificial ponds intended for fish production and wetlands to coniferous monocultures, heathland and spontaneous successional forests.

2.1.2. Brdy

The Brdy Highlands (Brdywald) is a rugged upland in central Bohemia. The area covers almost 700 km² of continuous forest at altitudes between 410 and 860 m a.s.l. The region is formed by the Bohemian Massif. The prevailing soil type is the cambisol (Chytrý 2012, Czech Geological Survey 2023). The highest points in the area contain Cambrian conglomerates and sandstones dominated by quartz. Because

of the infertile and acidic soils, the region remains heavily wooded and uninhabited, serving as a source of timber, charcoal, and ore processing activities (Žák, 2016). The climate in the region is moderately warm (Vondráková et al., 2013). The average air temperature during the breeding season is 7.2 °C (CHMI, 2023), and the Brdy Highlands receive less precipitation than other mountains in the Czech Republic (Žák, 2016). Approximately half of the forested area is a former military area, now designated as the Brdy Protected Landscape Area. The forests are mostly dominated by spruce, but the areas formerly used for military training have mostly been left to succession and are therefore covered by a diverse mixture of open habitats and young forests, often dominated by birch (AOPK 2015). We captured woodcocks at the Bahna site (49.7 N, 13.73 E).

2.1.3. Písecké hory mts.

The Písecké hory Mountains is a separate geomorphological unit of the Southern Bohemian Highlands. It consists of a continuous forested ridge between 410 and 630 m a.s.l. with an area of about 80 km². The region consists of the Bohemian Massif with cambisols of moderate productivity (Chytrý, 2012; Czech Geological Survey, 2023). It experiences moderate warmth (Vondráková et al., 2013) with a mean air temperature of 8.5 °C during the breeding season (CHMI, 2023). The forests are characterized by an extensive management regime in a protected nature park. Predominantly mature mixed forests are composed of a variety of tree species (beech, spruce, fir *Abies alba* and introduced Douglas fir *Pseudotsuga menziesii*), alternating with numerous species-rich plantations of different ages. At lower elevations near the edges of the forest complex, lime (*Tilia* spp.), maple (*Acer* spp.), alder (*Alnus glutinosa*) and willow (*Salix* spp.) are also common. With the exception of dense spruce monocultures where shrubs and herbs are mostly absent, most forests have rich undergrowth, especially in the herbaceous layer. We captured woodcocks in the central part of the area in a large and long-established clearing (49.31 N, 14.15 E).

2.2. Woodcock capturing and telemetry

We captured woodcock males during roding period on previously selected forest clearings with confirmed presence of roding males between early April and late June in 2021 and 2022. We roughly followed the method described by Heward et al. (2017), using mistnets, remote control playback and a decoy. Approximately 30 min before sunset, three 12 m long nylon mist-nets with a 30 × 30 mm mesh were erected as the open 'three-sided' box, with the fourth side formed by the forest edge. The playback lure and decoy were then placed near the box center. As Heward et al. (2017) recommended the use of a decoy with an emphasis on providing a moving stimulus, we used the plastic snipe with an additional tail made from light-colored domestic duck feathers, attached with a spring to an approximately 40 cm long stick. The decoy was then tied to a thin fishing line and moved by the observer hidden in the nearby bushes.

Another observer was hidden nearby to operate the playback device. As soon as the approaching male was heard or seen, the recording of the roding call was played (e.g. <https://www.xeno-canto.org/595410>). Unlike Heward et al. (2017), we used the combination of both the 'whistle' and 'grunt' components of the roding call. If the approaching male responded to the playback, e.g. by changing flight direction, we immediately changed the playback to the "chase contact call", which is commonly used in male-male confrontations (e.g. <https://www.xeno-canto.org/247609>). We started the capture effort at the beginning of the roding period (usually around sunset) and closed the nets either after a successful capture or 90 min after sunset.

We immediately removed the captured birds from the net and transported them in a canvas bag to a nearby quiet place for further processing. We marked the captured males with a metal ring, took basic biometric measurements and fitted them with Anitra GPS-GSM tags (ANITRA System s.r.o., <https://anitracking.com>, Backpack S - mini;

9–11 g). The tags were attached to the birds using a 4.76 mm PTFE (Polytetrafluoroethylene) bands and modified backpack method with a stretchable chest strap.

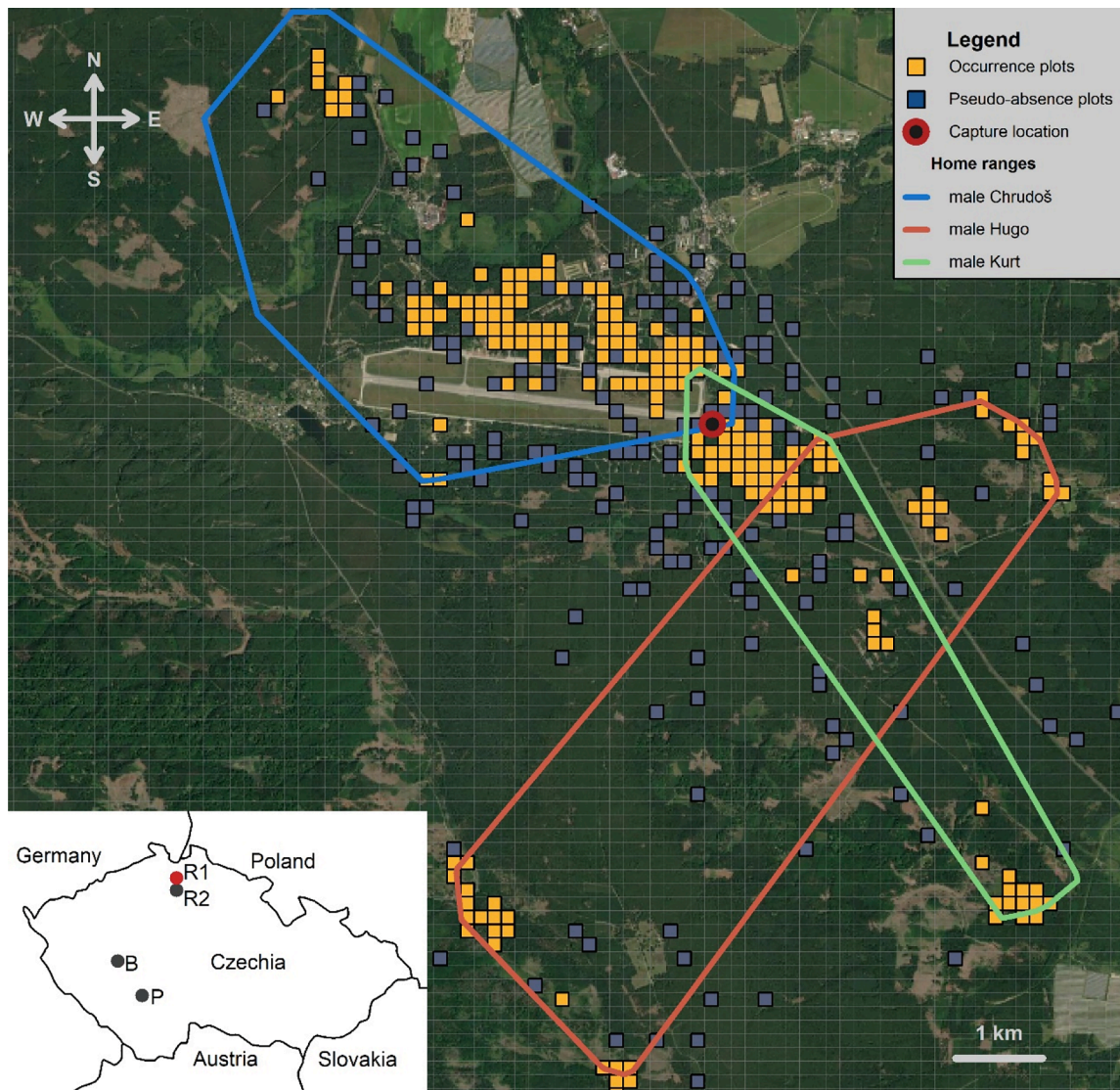
The tags we deployed are capable of recording and storing GPS position (typically to an accuracy in the low tens of meters, own unpublished data) together with multiple sensor readings (tri-axial accelerometry, tri-axial magnetometry, temperature, light levels and atmospheric pressure), typically once every few hours, and transmitting the data via the GSM network, typically once every two days if the GSM network is available. If not, the data is stored in the device's on-board memory and delivered when the tag reaches the GSM signal. The communication period was set to 9p.m., when we expected the bird to fly during the roding period. Depending on the battery level, the recording frequency was automatically adjusted between three and eight times a day. As the device is equipped with solar panels, the period of woodcock observation is theoretically lifelong. However, woodcock often spend time in dense vegetation with insufficient sunlight required by the solar panel and in places with poor GPS and/or GSM signal. As a result, the solar panel may have a poor battery charge and limited time to transmit data. Data coverage therefore varied considerably between individuals, often with long periods without providing the data.

2.3. Habitat data

2.3.1. Selection plots for the habitat survey

We conducted a field habitat survey to gain insight into habitat preference/avoidance by comparing the habitat used by woodcock males with the habitat available within their home ranges. Given the accuracy of GPS locations, we described habitats at the scale of hectare plots. Thus, all study areas were overlaid with the 100 m grid (Map 1, supplementary data on: <https://osf.io/j57mt/>). From the plots on the grid, we then selected 1) 'occurrence plots' where woodcock was recorded, and 2) the same number of 'pseudo-absence plots' that were randomly selected in the vicinity of the occurrence plots in presumed woodcock home ranges. As the habitats used by the woodcock males change considerably during the season due to vegetation growth, we categorized the season into three distinct phases and selected plots separately for each of them: "early season," spanning from the woodcock's arrival to May 15; "middle season," covering May 16 to June 30; and "late season," extending from July 1 until the individual's departure from the breeding area. Throughout the fieldwork conducted within these respective timeframes, we focused exclusively on plots visited by woodcock within the same season phase (and corresponding 'pseudo-absence' plots). While all field habitat surveys were conducted in 2022, we also include data from 2021. As the majority of the survey plots were forested, we assumed that habitat conditions within plots remained relatively stable from year to year, unless there was evidence of recent management intervention, such as new forest clearing. Two plots showing signs of recent management were excluded from the analysis. The selection of the plots for the study of the habitat during the different phases of the season was carried out as follows:

As 'occurrence plots' we selected all plots with at least one woodcock GPS position during the corresponding phase of the season. However, in order to select only those plots that were actually visited by woodcock, the raw GPS data had to be filtered in the following steps. Firstly, to avoid records taken in flight, we used only positions with a GPS-based speed estimate of less than 10 km/h. Secondly, to avoid data with a highly inaccurate GPS position, we didn't use locations where the difference between the altitude reported by the tag and the true altitude at the given coordinates was greater than 100 m, because our previous analyses showed that such records were usually also erroneous in the remaining coordinates (i.e., latitude and/or longitude). The true elevation at the given coordinates was estimated using the 'get_elev_point' function from the 'elevatr' R package (Hollister and Tarak 2017), with the 'z' argument set to 11, corresponding to the ground resolution of ~ 50 m. If the same plot was visited by either the same or different



Map 1. Map of the Czechia (lower left corner) with points marking the localities where woodcocks were studied from top to bottom: **R1** - Ralsko - Hradčany airport, **R2** - Ralsko - Vrchbělá, **B** - Brdy - Bahna and **P** - Písecké hory Mts. For the locality of the Ralsko - Hradčany airport (marked by red dot) the detailed situation is provided. The home ranges of the three woodcock males tracked at this site, taken as a 95% minimum convex polygon of all obtained locations, are shown as polygons. All three males were captured on the location marked by the red symbol with black center. The grid of 1-hectare plots used to select plots for the habitat survey is shown by a light gray net, and the surveyed plots are colored (occurrence plots are yellow and pseudo-absence plots are blue).

woodcock during the different phases of the season, it was surveyed repeatedly.

The 'pseudo-absence plots' were randomly selected according to the following algorithm. Using the 'mcp' function from the 'adehabitatHR' R package, we fitted minimum convex polygon around all GPS locations for the particular woodcock and phase of the season and extended this polygon by a 500 m buffer. We then selected a number of 1 ha plots equal to the number of corresponding occurrence plots (See Map 1). Unlike the occurrence plots (see above), the 'pseudo-absence' plots were only selected once. That is, we only selected plots that were not visited by other woodcock or that had already been selected as 'pseudo-absence' plots during the previous phase of the season. However, the effect of this rule on the results is obviously negligible, as only ~ 3 % of the potential 'pseudo-absence' plots were selected.

The status of some plots ('occurrence/pseudo-absence') changed later, as some 'pseudo-absence' plots were visited by woodcock shortly after our inspection. These plots were excluded from analyses. We were also unable to access some plots, mainly due to fencing of private land. As a result, the final number of woodcock occurrence plots and 'pseudo-

absence' plots differed slightly (342 occurrence plots and 323 'pseudo-absence' plots).

2.3.2. Habitat survey

All authors except EPV participated in the field surveys. During the field investigation, researcher briefly walked through the selected plot, and then completed the habitat questionnaire from its central point, using a Survey123 (<https://survey123.arcgis.com>) app on a smartphone. All habitat characteristics listed in Table 1 were estimated by eye as a percentage of the whole 100x100 m plot, usually to the nearest 5 %, but for marginal habitat categories to the nearest 1 %. To ensure consistency and similar perception of habitat proportions assessed by multiple researchers, we repeatedly trained by making several joint assessments of the same plot, followed by a discussion of differences in perception of certain variables and proportional estimates. All researchers were also provided with a detailed protocol defining all variables assessed. The researcher conducting the survey was unaware of the status of the plot visited (occurrence/pseudo-absence). The questionnaire also included taking a panoramic image for later reference.

Table 1
Habitat characteristics scored during the survey on selected plots and their description.

Variable	Description
Structure of the tree layer	
Canopy cover (%)	The proportion of forest floor covered by the vertical projection of the tree crowns
Structure of the forest floor cover (adds to 100 %)	
Bare soil (%)	The proportion of forest floor covered by the bare soil (most often areas rooted by wild boars, forest paths, or tree uprooting)
Leaves/needles (%)	The proportion of the forest floor covered by the leaves or needles
Vegetation (%)	The proportion of the forest floor covered by the plants including mosses
Water (%)	The proportion of the forest floor covered by the open water surface
Dead wood (%)	The proportion of the forest floor covered by the dead wood including tree stumps
Structural characteristics of vegetation	
Shrub shelter (%)	The proportion of forest floor, where the standing woodcock will not be visible from above, because of the shrubs and tree branches of low height (<1m)
Herb shelter (%)	The proportion of forest floor, where the woodcock may stand, but will not be visible from above, because of the herbs and ground-level shrubs (e.g. blueberries or heather)
Terrain traversability (%)	The proportion of forest floor that allows the woodcock to walk freely, i.e., the proportion of dense vegetation covering the plot (such as dense blueberry bushes), or other obstacles (such as dead wood) that might be difficult for a woodcock-sized bird to pass through was excluded.
Species composition of vegetation	
Deciduous trees (%)	The proportion of broadleaved trees within the tree layer
Dominant tree species	The most abundant tree species within the tree layer (i.e. not in the shrub layer)
Dominant herb species	Up to five herb species, which are dominant/important at the plot, together with their coverage (% of all herb vegetation)

2.4. Statistical analysis

2.4.1. General procedure of the statistical modelling

All statistical analyses were performed in R 4.0.2 (R-Core-Team 2022). All statistical models were constructed using the 'Hmsc' function and fitted using the 'sampleMCMC' function from the 'Hmsc' R package (Tikhonov et al. 2020). The models were sampled in two chains, each with 250 warm-up and 1000 sampling iterations, but with a different thinning factor (5–20) to achieve a potential scale reduction factor below 1.1, following the recommendations of Ovaskainen and Abrego (2020). As all models presented had a binary response, we used a probit link function. Estimates are presented as medians and 95 % credible intervals. All continuous predictors were z-transformed (mean centered and divided by standard deviation) (Schielzeth 2010). Whenever we distinguished between 'day' and 'night', the time was considered to be night when the sun was > 6° below the horizon. The position of the sun was calculated using the 'solarpos' function from the 'maptools' R package (Bivand and Lewin-Koh, 2020).

2.4.1.1. Home range size. We estimated home ranges in two different ways. First, we computed 95 % minimum convex polygons for all known positions of a particular woodcock during the particular breeding season, provided by the 'mcp' function from the 'adehabitatHR' R package (Calenge 2006). This method was used to compare the home range sizes with previously published estimates (Hoodles and Hirons 2007). However, home ranges estimated as minimum convex polygons often include large areas that are never used by the observed individual. Thus, we further calculated 95 % isopleths from local convex hulls (Getz et al. 2007), using the 'LoCoH.k' function from the 'adehabitatHR' R package (Calenge 2006), which was also previously used for the woodcock (Heward 2020). The 'k' parameter was set to the square root of the

number of GPS fixes used to calculate the home range (Table S1). Prior to calculation, GPS positions were filtered as described above. In addition, because some woodcocks made trips (often tens to more than a hundred kilometers) during the breeding season, we excluded trips longer than 10 km from the analysis, considering them rather as occasional trips outside the the home range boundaries that should not be considered part of the home range (Rolando 2002). As three of the woodcocks were tracked for two seasons, we calculated their home ranges separately for each year. As some GPS-GSM tags are only active for a very limited period of time, we only estimated home range size for those males that were tracked for at least 15 days during each season. For these males, home range size did not correlate with the number of days tracked ($r_s = 0.15$, $p = 0.63$), suggesting that we have approached the point of sampling saturation beyond which home range size does not increase significantly (Rolando 2002).

2.4.1.2. Movements within the home range. To describe the temporal distribution of movement within home ranges, we considered movement to be a change in position between the two successive GPS fixes if the position changed by at least 200 m. We first focused on exploratory visualization of the cases where we managed to obtain two GPS fixes within a short period of time (i.e. up to three hours). Since we were unable to determine the exact time of movement, we associated it with the midpoint between the two GPS fixes. Visualization of the recorded movements against the time of day (see supplementary scripts on) indicated that the probability of movement was low, except for the periods after dusk and before dawn. Therefore, we aimed to compare the probability of 'evening' and 'morning' movements. For the purpose of these analyses, we focused on the cases where we managed to obtain the GPS position during the night and the following day (i.e. potentially including the "morning movement") or during the day and the following night ("evening movement"). If more than one fix was obtained for a particular day or night, we took the distance between the last fix of the day (or night) and the earliest fix of the following night (or day), separated by at least three hours (to avoid spurious fixes around sunset or sunrise). We then estimated the probability of movement during morning vs. evening by fitting a binomial model with the probability of movement as the response, the type of movement ("morning" vs. "evening") as a predictor, and (as the probability of movement may be individually specific) an individual identity as a random effect.

2.4.1.3. General habitat preferences. To describe preferences for habitat structure parameters, we compared the plots which were visited by woodcock ('occurrence plots') with the randomly selected plots in the surroundings ('pseudo-absence plots' – see above). We wanted to compare habitat preferences between day and night separately, but some occurrence plots were visited by woodcocks during both day and night, so we fitted separate models for the day and night. The set of pseudo-absence plots was the same for both models ($n = 323$), while we defined 212 plots with occurrence during the day and 183 with occurrence during the night. Both models had a binomial response (woodcock presence: yes = 1, no = 0). We checked all potential predictors (listed in Table 1) for collinearity using Spearman's rank correlation. All variables had absolute values of the correlation coefficient < 0.5, except for the variables "Leaves/needles" and "Vegetation", which were strongly negatively correlated, i.e. almost complementary ($r_s = -0.81$). Of these two predictors we chose "Vegetation" because it is easier to assess by field inspection than leaf and needle cover. We also, we excluded variables that were zero in the majority of plots ('bare soil', 'water', 'dead wood'). The predictors finally included in the analysis were: canopy cover, deciduous trees, vegetation, terrain traversability, shrub shelter and herb shelter (for the exact specification of the variables see Table 1). In addition, to test for seasonal changes in habitat preferences, the day of the year that the plot was surveyed was included as a predictor, along with all of its interactions with habitat characteristics.

The random structure of the models included random effects (intercepts) of the locality (Ralsko, Brdy, Písecké hory Mts.) and bird ID. Plots visited by more than one male were entered multiple times (once for each visiting male). As habitat characteristics are typically spatially autocorrelated, we also controlled the habitat models for spatial autocorrelation by including plot center coordinates as a random effect.

The results of the models are presented as standardized effect sizes. Positive effects are reported as 'preference', while negative effects are commented as 'avoidance' of a particular habitat feature.

2.4.1.4. Preferences for dominant tree and herb species. A similar modeling approach was used to explore preferences for particular tree or herb species. For trees, we considered only a single dominant species (i.e. the most abundant) within the tree layer of the given plot, while for herbs, we considered all herb species that covered at least 20 % of the area within the given plot to be dominant. In some cases, we grouped several herb (or tree) species that were closely related and ecologically or structurally similar. Information on each modeled species or species group and sample size is summarized in [Tables S2](#) and [S3](#).

We modeled preference separately for each species (or group of species) that dominated at least 10 plots. In each model, the binary response was whether the plot was dominated by the particular species (yes = 1, no = 0) and the predictor was the status of the plot with respect to woodcock presence (pseudo-absence, occurrence – night, occurrence – day). To do this, we included plots with both day and night occurrences in both groups at the same time. We used the same random structure as for the models with general habitat characteristics, and we also present the same type of effect sizes. We also estimated and presented differences between all plot categories (pseudo-absence, occurrence – night, occurrence – day).

3. Results

3.1. Summary of sample sizes

A total of 13 males were tracked (7 captured in 2021, 6 in 2022). Three males captured in 2021 were tracked in both seasons ([Table S1](#)). In total, we obtained and analyzed 2262 GPS positions (after the filtering processes described above), covering the 342 occurrence plots. Within the subsequent positions, we defined 349 movements longer than 200 m.

3.2. Home range sizes and frequency of movements

The tracked males moved over a relatively large area within a single breeding season, typically several square kilometers. Their home ranges, estimated using a minimum convex polygon (MCP), ranged from 0.96 km² to 54.53 km² (median home range size was 7.26 km², $n = 13$, [Fig. S1a](#), [Table S1](#)). However, within the MCP, males typically used only a small portion of the area, repeatedly returning to similar locations ([Map 1](#)). Therefore, we estimated home range size using the local convex hull method. This approach reduced the area of home ranges to 0.19 km² – 6.49 km² and the median area to 1.01 km² ([Fig. S1b](#), [Table S1](#)), which represents 2 % – 89 % (median = 33 %) of the area over which males moved.

Woodcocks predominantly moved after sunset or before sunrise, with early morning movements more likely (model-estimated probability of movement = 78 %; 95 %CrI = 63 % – 87 %) than movements around sunset (48 %; 95 %CrI = 32 % – 63 %; [Fig. S1d](#)). When the birds changed position (i.e., when the two subsequent positions were > 200 m apart), the average movement distance between the two subsequent points was 1051 m (median = 837 m, 25th–75th quantile = 437–1392 m, max = 7019 m, $n = 349$; [Fig. S1c](#)), with evening movements marginally non-significantly longer than morning movements (Wilcoxon test, $w = 15414$, $p = 0.07$, $n = 129$ 'evening' and 214 'morning'

movements).

3.3. General habitat selection

While woodcocks generally spent majority of their time in diverse forest types (79.8 % from 2262 unique fixes; [Table S2](#)), they also visited other habitats, especially grasslands (16.5 %). To a lesser extent, woodcocks visited urban areas, rocky, arable, scrub and wetland habitats (all up to 1 %, [Table S1](#)). The use of non-woodland habitats was frequent especially during the night (27 % vs. 16 % during the day, [Table S2](#)).

When comparing the habitat characteristics of the plots where woodcocks were recorded with those of the pseudo-absence plots, woodcocks clearly preferred areas with good traversability, which also provided them with substantial amounts of shrub or herbaceous cover (or both; [Fig. 1](#), [Table 2](#) and [Table 3](#)). Interestingly, during the night woodcocks seemed to have an almost equal preference for both shrub and herbaceous cover, whereas during the day there was a clear preference for shrub cover ([Fig. 1](#), [Table 2](#) and [Table 3](#)). In addition, canopy cover had different effects on woodcock preference depending on the time of day. During the day they tended to stay in plots with denser canopy cover, while at night they preferred more open areas ([Fig. 1](#), [Table 2](#) and [Table 3](#)).

Throughout the breeding season, woodcock habitat preferences remained mostly stable. However, as the season progressed, there was a tendency for the preference for terrain traversability to decrease during the day, while it remained stable during the night ([Table 2](#)). Conversely, the preference for open areas during the night became even stronger as the breeding season progressed ([Table 3](#)).

3.4. Preferences for the tree species

Woodcocks inhabit a variety of forest types. The forests inhabited by woodcock males in our study were highly variable in terms of dominant tree species. Of the plots where woodcocks were present, 50.3 % were dominated by Scots pine, 25.6 % by birch, 11.7 % by spruce, and the remaining 12 % were distributed among other tree species/species groups ([Fig. S2](#), [Table S3](#), $n = 394$ one-hectare plots with at least one woodcock record). Compared to the randomly selected pseudo-absence plots, woodcock males highly preferred plots dominated by birch. During the night they also tended to slightly prefer plots dominated by fruit trees (mostly old orchards), although these plots were only marginally represented throughout the study area. Also, these plots were characterized by very sparse canopy cover ([Fig. S3](#)). Preference/avoidance for stands of other deciduous tree species remains unclear. On the other hand, plots dominated by conifers were avoided, even though they dominated to forested areas ([Fig. 2](#), [Table S5](#)). In particular, avoidance of pine-dominated sites was strong (although marginally non-significant) both during the day and at night, while spruce-dominated sites were avoided mainly at night.

3.5. Preferences for the herb species

In general, woodcocks did not appear to have a strong preference for (or avoidance of) high levels of plant cover ([Fig. 1](#)). However, a closer look shows that on the vegetation-covered plots, woodcocks preferred high grass (*Poaceae*) cover, including bush grass (*Calamagrostis epigejos*) ([Fig. 3](#), [S4](#), [Table S7](#), [S8](#)). On the other hand, they avoided growths of European blueberry (*Vaccinium myrtillus*) ([Fig. 3](#), [S4](#), [Table S7](#), [S8](#)). Furthermore, woodcocks slightly avoided plots with high moss cover, especially at night ([Fig. 3](#), [S4](#), [Tables S7](#), [S8](#)).

4. Discussion

Using high-precision GPS-GSM telemetry, we described the diurnal and nocturnal habitat selection of male Eurasian woodcocks throughout

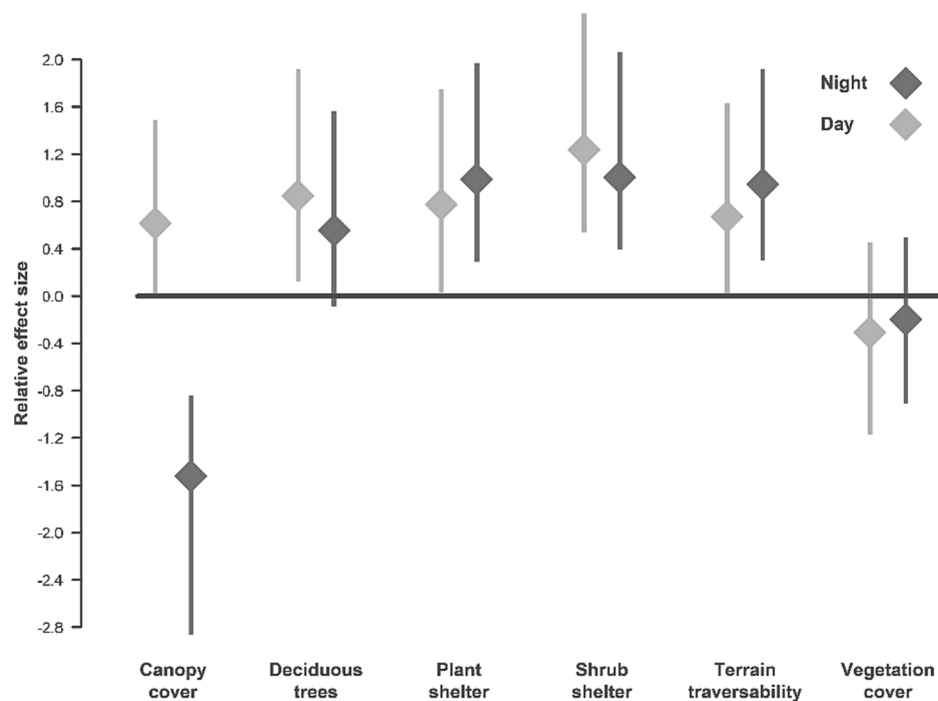


Fig. 1. Preferences of woodcock males for vegetation structure features during the day and night (sun > 6° below the horizon), compared to the randomly selected pseudo-absence plots. Points represent model-based estimates and vertical lines represent 95 % credible intervals (CrIs; Table 2, 3). A thick line indicates zero effect, positive values indicate preference of certain vegetation structure, while negative effect sizes indicate avoidance (i.e., negative preference). The CrIs that do not cross the zero line can be considered statistically significant.

the breeding season. Our data allowed us to elucidate several aspects of woodcock ecology in intensively managed Central European forests. We showed that (1) woodcocks have much larger home ranges during the breeding season than previously thought, (2) within their home ranges, woodcocks use many sites with highly diverse habitats, including intensively managed monocultures, although - compared to randomly selected pseudo-absence plots - woodcock males prefer deciduous (mainly birch) dominated forests. In general, they prefer sites with sufficiently dense vegetation cover but at the same time with good terrain traversability. Finally, (3) while woodcocks preferred plots with dense canopy cover during the day, they preferred more open plots at night, including various grasslands and other habitats. These aspects will be discussed in more detail below.

4.1. Home range size

The home ranges observed in our study are much larger than previously reported. While Hoodless and Hiron (2007) reported mean home ranges for Eurasian woodcock in Great Britain of approximately 60 ha, the mean home range estimated by a comparable approach (i.e. 95 % minimum convex polygon) in our study was twelve times larger (median 726 ha), and even the smallest home range was 94 ha. The home range size remains slightly larger even when estimated using the local convex hulls method, which is more sensitive to defining the area actually used by the tracked animal (Getz et al. 2007). These values are also relatively large compared to the breeding-season home ranges of the closely related American woodcock (*Scolopax minor*) (Hudgins et al. 1985), other shorebirds (Höglund and Robertson 1990, Johansson and Blomqvist 1996, Jonathan H. Plissner et al. 2000), or other forest birds of similar size (Redpath 1995, Rolstad and Rolstad 1995, Rolando 1998).

The first reason as to why home ranges we observed were larger could be the methodological differences between our study and Hoodless and Hiron (2007), as well as many above cited studies. In particular, the authors of the cited study used conventional radiotelemetry, a

method that involves actively searching for the tagged animal with a hand-held antenna, within the signal range of the transmitter, which can be much more limited in a forest environment than in open habitats. As a result, data may be lost if the monitored animal moves out of the range of the signal. However, we believe that this is unlikely to be the main cause of the observed difference, as the authors report a high frequency of woodcock position recording over a relatively long period of time (mean of 24 and 41 days for their two study areas, respectively). Such a period would be sufficient to detect the larger home ranges in our study, as the length of the tracking period and home range size became uncorrelated after only two weeks of tracking. We are therefore confident that the observed difference in home range size is not, at least to a substantial degree, a methodological artifact.

Second, several important drivers of home range size have been described for birds, with food availability, and foraging habitat fragmentation among the most important (reviewed in Rolando 2002). Although we consider it unlikely that forests in Central Europe are so much less prey productive than those in the UK that this could explain the more than tenfold increase in home range size, it is well known that the availability and abundance of earthworms (the main food of woodcock; Hoodless and Hiron 2007) in forests is highly variable in both space and time (Satchell 1983, Onrust and Piersma 2017, Martay and Pearce-higgins 2018). Woodcocks may therefore need to include a variety of habitats within their home range to ensure sufficient food supply throughout the breeding season and under different climatic conditions. Compared to the stable oceanic climate of the British Isles, this variability may be more pronounced in the more continental climate of Central Europe.

Finally, and perhaps most importantly, large home ranges may be a consequence of the size of the forest area. All three study areas consist of large and largely uninterrupted forest complexes covering tens to hundreds of square kilometers. This contrasts with much smaller and more fragmented forest areas that have been studied to date (Hiron 1980, Hoodless and Hiron 2007). Woodcocks are non-territorial (Hiron 1980), their home ranges often overlap (Map 1), and, as shown by

Table 2
Preferences of woodcock males for vegetation structure features during the day.

Response	Effect type	Effect	Estimate	Lower	Upper
Occurrence during the daytime (0/1)	fixed	Intercept	−1.571	−3.139	−0.033
		Date	0.047	−0.840	0.946
		Canopy cover	0.614	0.001	1.491
		Deciduous trees	0.848	0.123	1.921
		Vegetation cover	−0.305	−1.171	0.451
		Terrain traversability	0.672	−0.001	1.628
		Plant shelter	0.775	0.027	1.748
		Shrub shelter	1.237	0.541	2.399
		Date: Canopy cover	0.140	−0.602	0.972
		Date: Deciduous trees	0.400	−0.422	1.315
		Date: Vegetation cover	−0.256	−1.071	0.422
		Date: Terrain traversability	−1.148	−2.339	−0.437
		Date: Plant shelter	−0.777	−2.018	0.135
		Date: Shrub shelter	0.107	−0.592	0.868
		Male ID (Intercept)	1.0 %		
		Locality ID (Intercept)	4.2 %		
		Spatial autocorrelation	77.7 %		
	random				
	% variance				

The posterior estimates (medians) of effect sizes with 95% credible intervals (CrIs) from a posterior distribution of 1000 simulations generated by the “sampleMCMC” function from the Hmsc library (Tikhonov et al. 2020). A probit link function was used. Continuous predictors were z-transformed (mean centered and divided by SD). The effects whose 95% credible intervals do not include 0 are highlighted in bold.

Mulhauser and Santiago (2006), woodcock roding activity is not evenly distributed across large forest areas, but males tend to cluster around specific ‘roding centers’. And since woodcock males are not the caregiving sex during reproduction and the number of females available for mating varies considerably throughout the season, it can be expected that males may maximize their own fitness by visiting more such ‘roding centers’ over the large area (Kempenaers and Valcu 2017). In a large forest complex, there are likely to be more roding centers, probably separated by several kilometers of areas with little or no roding activity. Males living in such large forest complexes may therefore be motivated to visit more roding centers, for example based on the actual number of females ready to mate (Kempenaers and Valcu 2017). Switching between roding areas on different evenings has been proven repeatedly (Hirons 1980, Heward 2020). And because the evening roding period is relatively short and woodcocks are vulnerable to predation while flying during the day (Braña et al. 2010), it may be more advantageous to move to another roding site already during the night (either immediately after roding or before sunrise), spend the day there and be prepared for roding there the next evening, than to operate more distant roding sites from a small home range. Consistent with this hypothesis, tracked males repeatedly visited several widely separated (~several km) sites, which they revisited several times during the season, while avoiding large areas between these distant points (although with comparable habitat composition). In other words, the size of home ranges defined by minimum convex polygons was substantially larger than those defined by local convex hulls, with most occurrence plots located

Table 3
Preferences of woodcock males for vegetation structure features during the night.

Response	Effect type	Effect	Estimate	Lower	Upper
Occurrence during the night (0/1)	fixed	Intercept	−1.877	−3.084	−1.001
		Date	0.301	−0.405	1.138
		Canopy cover	−1.647	−2.811	−0.915
		Deciduous trees	0.556	−0.122	1.406
		Vegetation cover	−0.238	−1.031	0.518
		Terrain traversability	1.047	0.372	2.004
		Plant shelter	1.057	0.315	2.141
		Shrub shelter	1.152	0.536	2.167
		Date: Canopy cover	−0.758	−1.598	−0.129
		Date: Deciduous trees	0.008	−0.799	0.84
		Date: Vegetation cover	−0.495	−1.295	0.162
		Date: Terrain traversability	−0.22	−1.012	0.566
		Date: Plant shelter	0.098	−0.747	0.928
		Date: Shrub shelter	−0.209	−0.893	0.394
	random	Male ID (Intercept)	17.5 %		
	% variance	Locality ID (Intercept)	0.1 %		
		Spatial autocorrelation	64.2 %		

The posterior estimates (medians) of effect sizes with 95% credible intervals (CrIs) from a posterior distribution of 1000 simulations generated by the “sampleMCMC” function from the Hmsc library (Tikhonov et al. 2020). A probit link function was used. Continuous predictors were z-transformed (mean centered and divided by SD). The effects whose 95% credible intervals do not include 0 are highlighted in bold.

at their edges (see Map 1). However, further research, including monitoring of roding behavior together with the monitoring of female availability, will be needed to prove the proposed explanation.

4.2. Habitat preferences

Our data indicate that woodcocks prefer deciduous (and especially birch-dominated) forests, even though they represent a minority of the forests in the study areas. This is probably best explained by the higher abundance of earthworms, a key food item of the woodcocks, which has been shown elsewhere to influence woodcock habitat preferences (Hirons and Johnson 1986, Duriez and Corda, 2005, Hoodless and Hirons 2007, Braña et al. 2013). Although we did not directly assess earthworm abundance, our results fit well with reports, that earthworm abundance is negatively affected by low pH, and by high C:N ratios (De Wandeler et al., 2016), values typical of coniferous forests (Vesterdal et al. 2008, Cools et al. 2014). Earthworms are thus known to be much more abundant in deciduous or mixed forests than in coniferous monocultures (Satchell 1983). On the other hand, although pine and spruce forests were used less than would be expected by chance, areas dominated by conifers were still among frequently used habitats. Specifically, coniferous forests accounted for 60 % of the woodcock occurrence plots (compared to 70 % of the random plots; Table S2). Woodcocks are therefore likely to be willing to tolerate a very wide range of forest habitats. This may indicate that woodcocks are not motivated by a preference for specific tree species, but rather by specific microhabitat characteristics that meet their specific needs, such as safety and food.

Firstly, because woodcocks are not very agile and are therefore

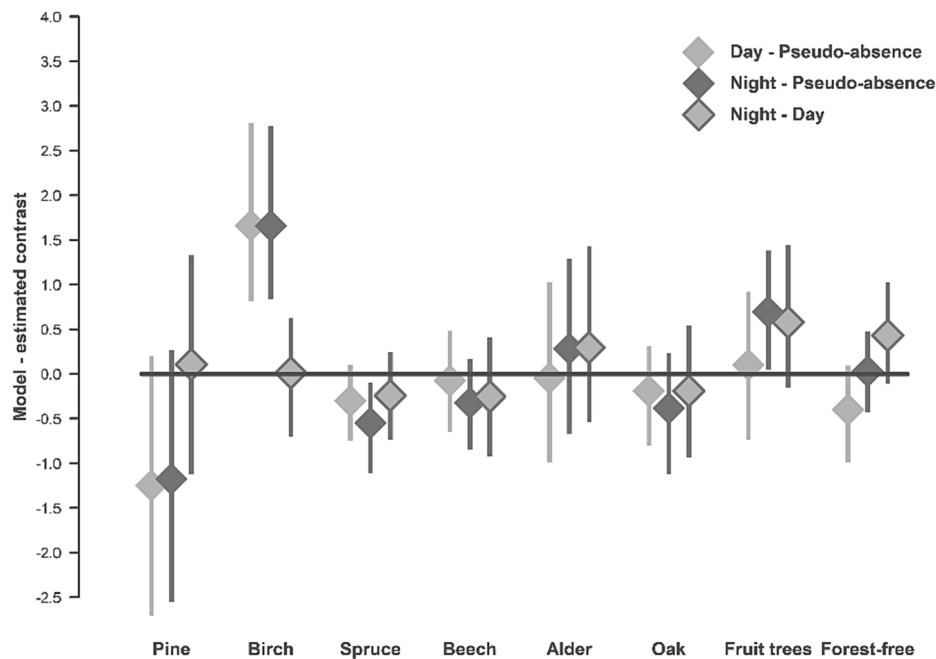


Fig. 2. Preferences for particular dominant tree species. Paired differences are shown between the day ($n = 212$) and night ($n = 183$) and between plots with woodcock occurrence and randomly selected pseudo-absence plots ($n = 323$). Points represent model-based estimates and vertical lines represent 95 % credible intervals (CrIs; Table S5, S6). A thick line indicates no effect, positive values indicate preference for a certain tree species, while negative effect sizes indicate avoidance (i.e., negative preference). The CrIs that do not cross the zero line can be considered statistically significant.

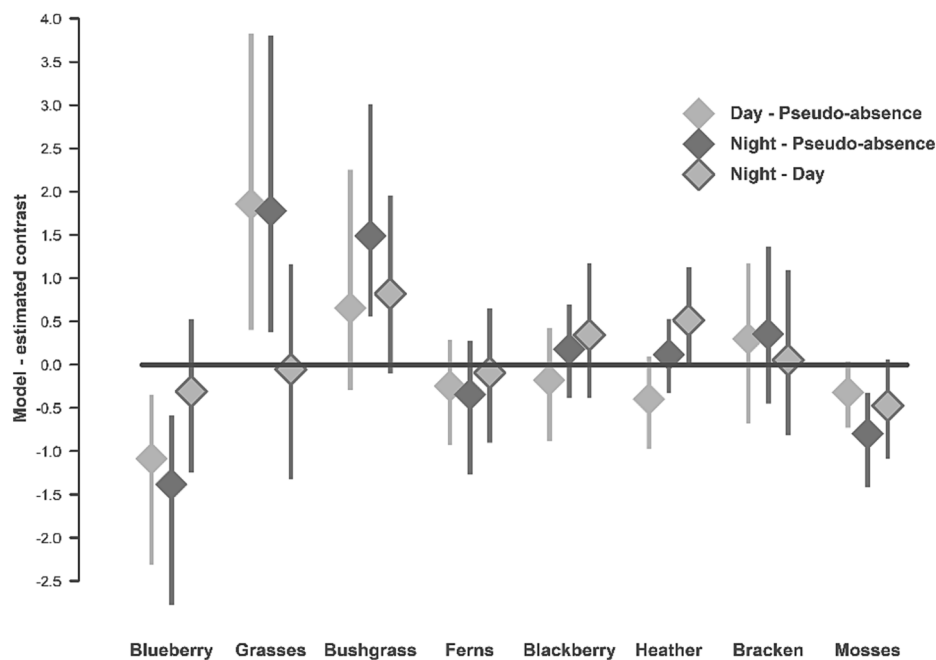


Fig. 3. Preference of woodcock males for specific herb species or groups of herb species (Table S4). Paired differences are shown between day ($n = 212$) and night ($n = 183$) and between plots with woodcock occurrence and randomly selected pseudo-absence plots ($n = 323$). Herb species are considered dominant if they cover more than 20 % of the plot. Points represent model-based estimates and vertical lines represent 95 % credible intervals (CrIs; Tables S7, S8). A thick line indicates zero effect, positive values indicate preference for a certain herb species, while negative effect sizes indicate avoidance (i.e. negative preference). The CrIs that do not cross the zero line can be considered statistically significant.

vulnerable to avian predators (Braña et al. 2010), they need to stay in places with sufficient overhead cover. In spruce and pine forests this is provided by a dense growth of young trees with branches reaching down to the ground. In deciduous forests, overhead cover may be provided by a different type of shrub, ferns, or other cover-providing plants. In previous studies, such habitat features have been represented by dog's

mercury and brambles (*Rubus* spp.) (Hoodless and Hiron 2007), or by very young tree stands with a canopy height of up to 2 m (Braña et al. 2013).

Secondly, woodcocks need to be able to move freely on the ground without obstacles when foraging. Hence, we showed that woodcocks preferred habitats with good terrain traversability (Fig. 3), while they

avoided plots covered with dense dead wood left after forest clearings and plots dominated by species with dense undergrowth, e.g. blueberry (*Vaccinium myrtillus*). In addition to free movement, such a preference may also allow woodcocks to detect approaching ground predators such as red fox (*Vulpes vulpes*) or martens (*Martes* spp.) well in advance. Despite the preference for good traversability, we also observed a preference for plots dominance by bushgrass, which is known for its dense growth and limited traversability. However, since our method does not allow us to pinpoint the exact location of the woodcock, this preference may be due to another habitat feature preference, as bushgrass is typically interspersed within various microhabitats that meet the woodcock's needs.

Thirdly, the search for earthworms is probably more efficient in places with shallow litter (Hirons and Johnson 1986), or even better in completely uncovered ground (Braña et al. 2013). This point of view may help us to understand the occasional presence of woodcock on arable land (Table S2), or the preference for birch and other deciduous forests with shallower leaf litter (Bray and Eville 1964), especially in spring and summer (Berg and Ryszard 2006). However, microhabitats with little or no litter can also be found in coniferous forests. In our study sites, these may be the forest roads, areas rooted by wild boar (*Sus scrofa*) or areas plowed as part of forest management prior to planting Scots pine. All of these microhabitat features were common in plots where woodcocks occurred. We can speculate that easy access to earthworms at these specific sites may outweigh the lower abundance of earthworms in continuous coniferous forests with the predominance of a compact needle cover on acidified soil (Satchell 1983).

4.3. Day-night differences

In line with previous studies (Duriez et al., 2005a, Hoodless and Hirons 2007, Braña et al. 2010), the main difference in habitat preference between day and night was that woodcock preferred to spend the night in more open habitats, which are probably more efficient from a foraging perspective. However, compared to these studies, we found a contrasting composition of such open habitats. Specifically, only 23.5 % of nocturnal fixes were on grassland or arable land (Table S2; compared to the 76 % reported during the breeding season by Hoodless and Hirons 2007), and only 12 plots with nocturnal woodcock occurrence (i.e. 6 %) had zero canopy cover. Note also that none of the plots were located on pastures, which are often reported elsewhere (Duriez et al., 2005a, Hoodless and Hirons 2007, Braña et al. 2010). Thus, most of the 'open habitats' in our study area were open forest habitats, such as forest clearings, newly planted sites, or simply the sites with less dense canopy cover (e.g. around wind uprooted trees). In addition, plots with spontaneous succession or old orchards were represented. The preference for open areas within the forest rather than the open agricultural areas may be due to a preference for staying in a safe habitat with sufficient cover, even at night. This is also indicated by the fact that the importance of plant shelter increased more during the night than during the day, while the importance of shrub shelter remained similar (Fig. 3). As predation risk for woodcock has been shown to be higher in open habitats (Duriez et al., 2005a), the availability of suitable foraging habitats (i.e. patches with looser canopy cover) within intensively managed forests may be an important positive feature of Central European forests that supports the occurrence of the woodcock. On the other hand, it is possible that woodcocks occupying large forest areas are simply forced to seek out sites with sparse canopy cover within the forest because of the scarcity of other habitats (e.g. pastures) in their home range (e.g. compared to the UK).

While the habitat preferences described above remained largely consistent throughout the season, we found two exceptions. Firstly, in contrast to the unchanging preference for open habitats reported for the British population (Hoodless and Hirons 2007), the nocturnal preference of woodcock for more open sites (i.e. plots with less continuous canopy cover) increased. Secondly, the diurnal preference for microhabitats

with high terrain traversability decreased. A likely explanation for both of these trends is the aforementioned need for well-traversable foraging sites with easily accessible earthworms and, at the same time, cover from above. As vegetation grows during the season, it may become more difficult to find a site that meets all of these requirements at the same time. Therefore, during the day, woodcocks may prefer to stay in the safer habitat (although perhaps not with abundant and/or easily accessible earthworms), while at night they tend to compensate it by visiting more open areas with readily available food. Whether this is accompanied by a change in the circadian rhythmicity of foraging behavior and food intake requires further research.

4.4. Potential effect of male biased sampling

The limitation of this study was that we could only describe the behavior of males. Unfortunately, it is much more difficult to find or even capture female woodcocks during the breeding season (Heward et al. 2017). Consequently, a similar problem of male bias in telemetry data used to describe woodcock behavior during the breeding season is common to other studies (Hirons 1980, Hoodless and Hirons 2007, Braña et al. 2013).

Since females are the only parental caregivers, we cannot exclude that the habitat requirements of females during incubation or chick rearing may be different and/or more specific. At the very least, we can expect females to have significantly smaller home ranges during the breeding season. However, we believe that the general habitat requirements are likely to be similar, particularly for the following reasons. Firstly, Hoodless and Hirons (2007) found no significant differences in habitat preference between the sexes, even though females were a large minority in their sample. Secondly, the known (albeit sporadic) descriptions of microhabitat preferences during nest placement (Hirons and Johnson 1986, Nutilová 2023) indicate that woodcocks may place their nests in extremely variable forest habitats (Nutilová 2023), with a possible preference for sites sheltered by plants, namely bramble (Hirons and Johnson 1986), or deciduous trees, namely birch (Nutilová 2023). However, the study of female behavior, including the methodological aspect of capturing females during the breeding season and an efficient technique for searching for nests, remains an important gap and a key task for future research.

5. Conclusions

In Central European forests, woodcocks move over relatively large areas, with home ranges typically covering several to tens of square kilometers. This allows them to choose from a relatively wide range of different, mainly forest, habitats. Woodcocks can be found in intensively managed coniferous forests but do not prefer them. Although they prefer stands of several broadleaved species (mainly birch) and tend to avoid coniferous forests, they appear to be able to find suitable sites even in intensively managed conifer monocultures. Their site selection decisions are therefore likely to be influenced by forest structure rather than tree species composition. Specifically, they use sites with a dense but not excessive cover of shrubs and/or plants, which allows them to move freely while foraging. At night, they prefer diverse, more open habitats, but usually within the sparse forests while less often in the country outside. Therefore, both good protection from predators in the form of tree, shrub, and herbaceous layers, especially during the day, and sufficient food (earthworms) available at night, are important conditions for supporting the occurrence of woodcock in the commercial forests of Central Europe.

From a forest management perspective, our data emphasize the need to establish and maintain forests that are species- and especially structurally diverse. Several recommendations can be made to increase the suitability of forests for woodcock and to support the Central European woodcock population. These include increasing the amount (proportion) of small patches dominated by commercially non-target

broadleaved trees, namely birch, to allow the development of structurally diverse shrub (or plant) cover and to improve soil quality, which is important for abundant earthworms and woodcock foraging. This can be achieved by increasing the proportion of sites left to spontaneous succession over a longer period (e.g., several decades), e.g. less suitable sites for planting such as wet clearings. In addition, areas without excessive impassable ground vegetation, but with sparser canopy cover and denser understory herbaceous and shrub vegetation, play an important role in providing reliable cover for woodcock, especially during the day. Such areas should be more strongly supported, for example through appropriate management practices such as forest thinning or natural regeneration from seed. In addition, small patches of dense tree planting adjacent to sparse mature forests can increase shelter as well as availability of food for woodcock in an area. A combination of such specific microhabitats interspersed within continuous coniferous production forests could significantly support woodcock populations. We expect that such changes would also have a positive impact on other forest species and thus benefit the biodiversity of commercial forests in Central Europe.

CRedit authorship contribution statement

Martin Sládeček: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization, Project administration. **Lucie Pešková:** Investigation, Resources, Data curation, Writing – review & editing, Project administration, Funding acquisition. **Petr Chajma:** Methodology, Investigation, Writing – review & editing. **Kateřina Brynychová:** Investigation, Writing – review & editing. **Kateřina Koloušková:** Investigation, Data curation. **Kateřina Trejbalová:** Investigation, Writing – review & editing. **Veronika Kolečková:** Investigation. **Eva Petrusová Vozabulová:** Data curation, Writing – review & editing. **Miroslav E. Šálek:** Conceptualization, Investigation, Writing – review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared the link to our data and scripts placed on the OSF repository (<https://osf.io/j57mt/>)

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Ethics declaration

We confirm that all animal experiments conducted for this study adhered to the guidelines stipulated by the ARRIVE guidelines, U.K. Animals (Scientific Procedures) Act, 1986, EU Directive 2010/63/EU, and the NIH guide for the care and use of Laboratory animals. Also, the study was conducted in accordance with the current laws of the Czech Republic and European guidelines. All co-authors who handled birds have Ringing Permits.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2023.121489>.

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